

4.10.18: Area of a Bounded Region

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Question

Find the area of the region bounded by the lines $y = 4x + 5$, $y = 5 - x$, and $4y = x + 5$.

Step 1: Finding the Vertices

Vertex A: Intersection of $y = 4x + 5$ and $y = 5 - x$

$$4x + 5 = 5 - x \implies 5x = 0 \implies x = 0, y = 5$$

$$\mathbf{A} = \begin{pmatrix} 0 \\ 5 \end{pmatrix}$$

Vertex B: Intersection of $y = 5 - x$ and $4y = x + 5$

$$4(5 - x) = x + 5 \implies 20 - 4x = x + 5 \implies x = 3, y = 2$$

$$\mathbf{B} = \begin{pmatrix} 3 \\ 2 \end{pmatrix}$$

Vertex C: Intersection of $y = 4x + 5$ and $4y = x + 5$

$$4(4x + 5) = x + 5 \implies 16x + 20 = x + 5 \implies x = -1, y = 1$$

$$\mathbf{C} = \begin{pmatrix} -1 \\ 1 \end{pmatrix}$$

Step 2: Calculating the Area

We use the determinant formula for the area:

$$\text{Area} = \frac{1}{2} |\det(\mathbf{A} - \mathbf{B}, \mathbf{A} - \mathbf{C})|$$

First, we find the side vectors:

$$\mathbf{A} - \mathbf{B} = \begin{pmatrix} 0 \\ 5 \end{pmatrix} - \begin{pmatrix} 3 \\ 2 \end{pmatrix} = \begin{pmatrix} -3 \\ 3 \end{pmatrix} \quad (1)$$

$$\mathbf{A} - \mathbf{C} = \begin{pmatrix} 0 \\ 5 \end{pmatrix} - \begin{pmatrix} -1 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 4 \end{pmatrix} \quad (2)$$

Step 2: Calculating the Area (continued)

Now, we calculate the determinant:

$$\begin{aligned}\det(\mathbf{A} - \mathbf{B}, \mathbf{A} - \mathbf{C}) &= \begin{vmatrix} -3 & 1 \\ 3 & 4 \end{vmatrix} \\ &= (-3)(4) - (1)(3) \\ &= -12 - 3 = -15\end{aligned}\tag{3}$$

The area is half the absolute value of the determinant:

$$\text{Area} = \frac{1}{2}|-15| = 7.5\tag{4}$$

The area of the triangle is 7.5 square units.

C Code - Verification

```
#include <stdio.h>
#include <stdlib.h>
#include <math.h> // For fabs

double triangle_area() {
    double Ax = 0, Ay = 5;
    double Bx = 3, By = 2;
    double Cx = -1, Cy = 1;
    // Area formula using determinant concept
    double area = 0.5 * ((Ax - Cx)*(By - Ay) -
                        (Ax - Bx)*(Cy - Ay));
    return fabs(area);
}

int main() {
    double area = triangle_area();
    printf("Area of triangle = %.4f\n", area);
    return 0;
}
```

```
import ctypes
import numpy as np
import matplotlib.pyplot as plt

# Load C shared library
lib = ctypes.CDLL('./libtriangle.so')
lib.triangle_area.restype = ctypes.c_double
area = lib.triangle_area()
print(f"Area of triangle = {area:.2f}")

# Triangle vertices
A = np.array([0, 5])
B = np.array([3, 2])
C = np.array([-1, 1])
triangle = np.array([A, B, C, A])
```

Plot triangle

```
plt.plot(triangle[:,0], triangle[:,1], 'b-o', label='Triangle')
plt.fill(triangle[:,0], triangle[:,1], color='cyan', alpha=0.3)
```

Plot lines

```
x_vals = np.linspace(-3, 4, 400)
plt.plot(x_vals, 4*x_vals+5, 'r--', label='$y=4x+5$')
plt.plot(x_vals, 5-x_vals, 'g--', label='$y=5-x$')
plt.plot(x_vals, (x_vals+5)/4, 'm--', label='$4y=x+5$')
```

Annotate vertices

```
labels = ['A', 'B', 'C']
coords = [A, B, C]
for label, coord in zip(labels, coords):
    plt.annotate(label, (coord[0], coord[1]), textcoords="offset
        points",
        xytext=(10, -10), ha='center', fontsize=12, color
            ='black')
```



```
# Decorations
plt.legend()
plt.grid(True)
plt.axis('equal')
plt.xlabel('x')
plt.ylabel('y')
plt.title(f"Triangle Bounded by Lines, Area = {area:.2f}")
plt.savefig("triangle_ctypes_with_lines.png", dpi=200)
plt.show()
```

Pure Python Code

```
import numpy as np
import matplotlib.pyplot as plt

# Triangle vertices
A = np.array([0, 5])
B = np.array([3, 2])
C = np.array([-1, 1])
triangle = np.array([A, B, C, A])

# Area calculation
area = 0.5 * abs(np.linalg.det(np.column_stack((A-B, A-C))))
print(f"Area of triangle = {area:.2f}")

# Plot triangle
plt.plot(triangle[:,0], triangle[:,1], 'b-', linewidth=2)
plt.fill(triangle[:,0], triangle[:,1], color='cyan', alpha=0.3)
```

Pure Python Code

```
# Highlight vertices with larger markers
plt.scatter([A[0], B[0], C[0]], [A[1], B[1], C[1]],
            color='red', s=100, zorder=5, label='Vertices')

# Annotate vertices
labels = ['A', 'B', 'C']
coords = [A, B, C]
for label, coord in zip(labels, coords):
    plt.annotate(label, (coord[0], coord[1]), textcoords="offset
                  points",
                 xytext=(10, 10), ha='center', fontsize=12, color=
                 'black', fontweight='bold')

# Plot lines
x_vals = np.linspace(-2, 4, 400)
plt.plot(x_vals, 4*x_vals + 5, 'r--', linewidth=1.5)
plt.plot(x_vals, 5 - x_vals, 'g--', linewidth=1.5)
plt.plot(x_vals, (x_vals + 5)/4, 'm--', linewidth=1.5)
```

Pure Python Code

```
# Annotate line equations near the triangle
plt.text(-1.8, 5.2, '$y=4x+5$', color='r', fontsize=12,
         fontweight='bold')
plt.text(2.1, 3.5, '$y=5-x$', color='g', fontsize=12, fontweight=
         'bold')
plt.text(0.5, 1.5, '$4y=x+5$', color='m', fontsize=12, fontweight
         ='bold')

# Zoom limits
plt.xlim(-2, 4)
plt.ylim(0, 6)

# Decorations
plt.grid(True)
plt.axis('equal')
plt.xlabel('x')
plt.ylabel('y')
plt.title(f"Triangle Bounded by Lines, Area = {area:.2f}")
plt.legend()
```

Plot of the Bounded Region

